# Saathi

Kaavish Report
presented to the academic faculty
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# Saathi

This Kaavish project was supervised by:

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Approved by the Faculty of Computer Science on ____________________.

# Dedication

This project is wholeheartedly dedicated to our beloved parents who have raised and nurtured us to become the individuals we are today. A special tribute to our parents for their endless love, sacrifices, prayers, and advice. We are nothing without your support. To our siblings and friends, who gave us advice, support, and inspiration to complete this endeavour and bolster us when we were struggling.

We would also like to dedicate this project to all the elderly people in our life who have inspired us to work on Saathi and create a solution with features that would hopefully make their everyday lives easier.

Most importantly, the project is dedicated to the spirit and perseverance of our group Saathi, where everyone sincerely supported and encouraged each other during tough times and spent countless hours of overtime work to ensure the project's success.

Lastly, we dedicate this project to the Almighty God, thank you for the guidance, strength, power of the mind, education, and skills, and for giving us a healthy life in these testing times of the pandemic. All our success, we owe to you.

# Acknowledgements

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# Abstract

With the rise of the digital age, life has become a lot easier for the vast majority of the population. However, the ever-increasing elderly population has suffered, especially in countries like Pakistan, where limited accessibility to technology, often due to language barriers, hinders elderly from reaping technological benefits. Saathi is an Urdu virtual assistant application which provides an intuitive and empathetic platform for the elderly in Pakistan. It helps them perform essential tasks such as reminding them of their medications, organising their work, getting daily news highlights, and connecting them with their loved ones. It also provides them with entertainment in the form of user-specified video playlists or by positively engaging them in conversations on various topics. For this purpose, two forms of conversation generation have been explored. Firstly, the open-source Rasa Framework is used for a custom domain-specific conversational agent. Secondly, this work also investigates the degree to which the power of fine-tuning can be harnessed to utilize pre-trained language models for open-ended natural language generation in Urdu.

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# 1. Introduction

## 1.1 Problem Statement

According to the World Health Organization (WHO), life expectancy is increasing due to advancement in science, strong economies, and healthy behaviors [15]. The latest census of Pakistan states that the current population of people above 60 (old-age) is estimated to be 5.54% [46]. Ageing Asia estimates that Pakistan's elderly population would grow to 12% of the total by 2050, translating to an additional 40 million individuals [4]. However, due to the increasing normalization of a nuclear family structure in Pakistani society [60] and busy lifestyles of modern day individual, oftentimes there are not enough people to take care of older family members and relatives. The emergence of the COVID-19 pandemic has also compelled the elderly to remain isolated and confined to their homes, oftentimes alone. This could lead to inconsistencies in care and diseases going undiagnosed, affecting both their emotional and physical health.

Most of the health issues that the elderly face are said to be linked to chronic conditions, particularly those that are non-communicable. By instilling healthy behaviours and providing a supportive environment, the majority of these illnesses can be avoided or slowed down. As a result, we believe it is an issue that deserves immediate attention, particularly in our local context, where progress has been slow.

Moreover, currently there is a lack of Urdu conversational agents and more so voice-based Urdu conversational agents. This lacking has caused a disconnect between mainstream technology and the Urdu speaking population. Lack of such agents cause the elderly to feel less connected and seen. Consequently, there is much needed work to be done in Urdu conversation using both existing frameworks as well as natural language generation through deep learning methods.

## 1.2 Proposed Solution

Through our project, we aim to provide elderly with a virtual assistant in the form of an Urdu chatbot. The chatbot aims to generate human-like conversations with the elderly and help them in executing their daily life tasks. This conversation generation has been performed through two channels: Rasa Framework and transfer learning of the pre-trained deep learning language models.

Our mobile application consists of basic functionalities, i.e. voice based mobile actions such as making calls and sending messages, playing predefined playlists, giving medicine reminders, alerting the elderly to charge their phones if low battery is detected and more. The voice bot also greets them, prompts them to take their meals on time, and generates conversation or actions accordingly (e.g., if the elderly person is feeling sad or unwell, it can suggest that they call a loved one). Additionally, we be send updates to the caretaker (via WhatsApp) if there is an emergency situation or if the elderly person needs to alert the care-taker in any scenario, e.g. in case their medicine runs out.

Since our app is voice-enabled, people with vision impairment, and limited mobility will also be able use it. The purpose of exploring different ways of conversation generation is to be able to respond to the user in a more sympathetic and personalised way. For our conversational agent, we have explored two main avenues. Firstly, we have used Rasa Framework for domain specific conversations. For this purpose we use natural language processing extensively in order to understand and process the user input in Urdu through speech and take actions accordingly. Secondly, we investigate the potential to which transfer learning can be harnessed for conversation generation in Urdu. For this we will be using three state-of-the-art transformer-based models which include DialoGPT [67] and two multilingual models mT5[65] and the XGLM [36] models which would be fine-tuned on our compiled UrduSaathi dataset.

**All in all, our project aims to create a mobile Urdu virtual companion for elderly people in Pakistan that facilitates them in their daily tasks and provides them with some form of support.**

## 1.3 Intended User

The intended user of our application are Urdu-speaking adults, both male and female, above the age of 60. They will mostly belong to the middle or upper middle class, be able to understand and converse in Urdu and have access to a personal smartphone.

## 1.4 Project gantt chart and deliverables

### Deliverables

1. Conduct interviews and survey to better asses the problems and requirements in order to refine our solution.

2. A Project Proposal that states the aims, objectives and goals of the team.

3. Literature review for different aspects of our project including but not limited to multilingual chatbots, open-domain conversation generation through deep learning models etc.

4. SRS documentation providing the details of the application and project.

5. SDS documentation listing the design details and how the application is built.

6. External interface of our application, that will showcase how the end-product works as a whole.

7. Chatbot development to build a domain specific Urdu chatbot in RASA

8. Dataset development: Urdu dataset had to be pre-processed and compiled for transfer learning for Urdu conversation generation

9. Database development for saving the necessary user data.

10. Urdu conversation generation using transferred learning

11. Integration of different components

12. Submission of the application prototype: Testing of integrated prototype for fixing any bugs

13. Working on research papers in order to publish our findings

14. Final report: compiling all the development and research work into a comprehensive report

# Gantt Chart

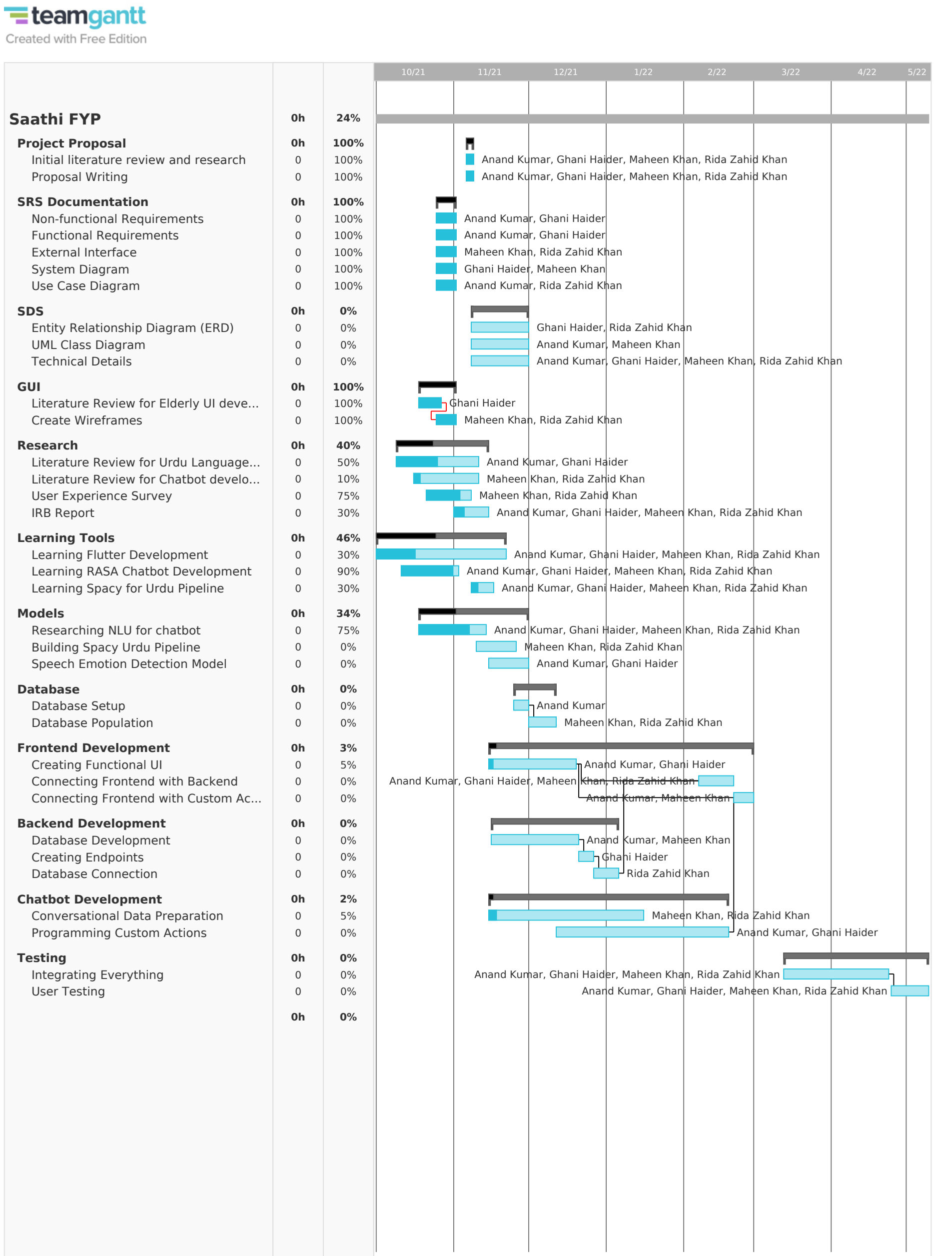



Figure 1.1: Saathi Gantt Chart

## 1.5 Key Challenges

### 1.5.1 Saathi Application

A key challenge for our solution was building the dataset for our RASA chatbot which would contain all the intents, entities and actions in the context of the problem, which is, providing assistance to the elderly.

We had to incorporate an NLP pipeline for Urdu in RASA in order for it to correctly extract intents and entities from the user text. For that, we used the SpaCy library to build the different components of the NLP pipeline. This also required a large Urdu text corpus for training our pipeline. There are other numerous existing challenges in this as well, such as the major hurdle in tokenization by the improper use of space between words in Urdu as well as the absence of case discrimination making the sentence boundary detection a difficult task [51]. In order to overcome these difficulties an extensive research on Urdu grammar and Urdu Language Processing was needed.

Another important component of our project is the speech-to-text and text-to-speech libraries that convert Urdu speech into text for processing of the user query and to respond back to the user in Urdu speech. Several libraries are available for this task, but most of them require paid subscriptions, therefore there was a need to further look for open source libraries which provide these functions with good accuracy and performance. We finally settled with Flutter speech-to-text and text-to-speech plugins after considering all the mentioned aspects

### 1.5.2 Conversation Generation

Building open-domain conversation systems that produce human-like and contextual responses is a recognized challenge[27]. Recent state-of-the-art transformer-based models like DialoGPT[67] have given promising results for simulating human-like single-turn conversations in English. However, to our knowledge, such models have not yet been used for the purpose of Urdu conversation generation. Conversation generation is a complex task that requires ample amounts of data similar to other generative tasks. Since Urdu is a resource-poor language, large open-source datasets are not currently available. This has been a significant reason behind the lack of work in Urdu conversation generation.

# 2. Literature Review

## 2.1 Elderly Care and Technology

The society, according to the United Nations, is experiencing a great demographic change known as ageing society. 13% of the world's population is now elderly (i.e. over the age of sixty), and this percentage is predicted to increase to approximately 25% by 2050 [63]. This poses several hurdles in the delivery of high-quality health and social care and may have significant impact on several socio-economic issues like the family sustainability and the economic growth. Ageing is a detriment to the person's health and, consequently, their autonomy. This dependency may force older adults to stay at home, which could lead to social isolation, feelings of depression or even greater dependency in completing everyday tasks [62]. This is why today, there is a growing demand for technology supporting the elderly at home.[16]

Much research and development has been done in the domain of Socially Assistive Robotics(SAR) to develop companion robots in order to assist older people in a variety of tasks such as monitoring and promoting physical health in the elderly at home, [16],[49], helping old people by autonomously navigating the user's home, reminding them of their medications, and providing entertainment [26]. Other than that, technologies such as Internet of Things (IoT) and wearable technologies could offer promising solutions for elderly care. Wearable devices are a part of IoT systems which can be worn by the people for monitoring the physical activities and physiological data [58]. These devices are embedded with sensors and algorithms that allow them to track, analyse and guide users' behaviour [55], vital signs or movement. Wearable tech has the potential to assist with several scenarios in elderly healthcare and can improve the quality of life for the elderly and allow them to maintain their independent lifestyle [58].

However, one of the major issues with SAR systems and wearable technology is the lack of social component and interaction, which hinders their acceptance among older people. Another problem with these assistive robots is their significant cost, which is not within everyone's budget, especially in low income countries like Pakistan. Unlike SAR, our goal is to provide the elderly with a low-cost virtual assistant in the palm of their hands via their cellphones, which would be both cost-effective for them and provide them with the necessary assistance.

## 2.2 Virtual Assistants

Like other technological devices, virtual assistants have become an element of great help to integrate and keep people active, in addition to facilitating the different everyday tasks that need to be performed [24]. A virtual assistant is an application that recognises voice instructions and does tasks on the user's behalf. Intelligent Virtual Assistants such as Amazon's Alexa Echo Dot, Apple's Siri, Google Assistant, Microsoft's Cortana, and Samsung's Bixby, among others, are commercially available and have a high level of public acceptability. [38].

Virtual assistants have a number of advantages when it comes to improving the quality of life for the elderly. These assistants serve as a means of providing social support and managing loneliness of elderly people due to their functionalities of being able to talk and respond [14] to the user and even engaging the user through games, playing music or even reading out the latest news. These assistants also help elderly people in staying connected with their friends and the community[53]. Through active listening skills and even acting to improve the user's mood, virtual assistants can help reduce symptoms of loneliness, boredom, despair, and irritation in the elderly [12].

There has been significant work done to develop user-centered virtual assistants that feature different functionalities to assist elderly people. Researchers have found that spoken language seems to be the most preferred mode of interaction for elderly people, they prefer to interact with virtual assistants in a simple, hassle free way [57]. Beskow et al [10] designed a user-centric virtual assistant that could be used to set medicine reminders after surgery and had a virtual character with whom users could communicate and set reminders, among other things. Lesin [34] developed a virtual assistant solely for mobile platforms, which aimed to assist diabetic patients with their daily routines. In addition to providing reminders, it also possessed the option to reach out to doctors/medical staff. Bickmore et al.[11] developed a virtual assistant to foster fitness and exercises in elderly people. Mival et al.[20] developed a virtual assistant that resembled a dog so that people may have virtual pets and communicate with them. Klein [32] created a virtual assistant that could aid a user in managing stress and anxiety by offering a variety of solutions for reducing stress and anxiety. Kasap et al. [30] created a virtual assistant that could change its facial expressions depending on how the user was feeling. Yasuda [66] created a virtual assistant with a human-like aspect to encourage user attention in a purely conversational medium.

The aforementioned works are in English and hence would not be helpful in the local context of Pakistan as majority of the population does not converse in English. Little work has been done in the field of virtual assistants in Urdu. The world's first "AI Urdu voice-recognition bot" called RUBA was presented at the Smart CX conference in 2019. RUBA stands for Real Urdu Bot Automation and is a Siri-styled virtual voice-enabled personal assistant. The bot can converse with the user in Urdu and react in Urdu while performing simple activities like as checking the amount of the user's bank account, sending text messages, and so on[64]. Other applications like *Ok Ameer* [42] are also available on the PlayStore however they do not necessarily provide the functionalities needed for the ease of use of the elderly.

## 2.3 Chatbots in Urdu

While researching for chatbots, we were able to find many that were in English and other mainstream languages, but limited work has been done in the mainstream for non-english languages including Urdu. However, UMAIR (Urdu Machine for Artificial Intelligent Recourse) is a conversational agent in Urdu developed as a customer service representative [37]. The authors discuss several challenges that one might face while creating a chatbot/conversational agent in Urdu. Urdu as a language, might be easy to understand for a human being who knows the language. However, Urdu, like Arabic uses diacritics (marks) above and below the alphabets which makes it difficult for a machine or any kind of computational system to understand text without the presence of these diacritics. Another limitation when it comes to Urdu is related to its morphology. Urdu does not have spaces between characters, nor does it make use of capital and small alphabets to indicate the use of a proper noun. All of this constitutes to making Urdu a fairly complex language. As a result, proper word segmentation has to be performed before strings can be processed further for the purpose of intent and entity recognition. [37]

However, there still exist some Urdu chatbots that are currently being used for several purposes. Raaji by Aurat Raaj is an Urdu chatbot created to help women of all ages with questions related to menstrual hygiene and reproductive health. [9] Another Urdu/Telugu/English chatbot was launched by the Telangana government in India during the pandemic to provide authentic information related to Covid-19 to the citizens. This chatbot does not necessarily have a conversation element, rather it is menu-based and can only be used to provide information to the users. [61] Similarly, in 2020, the government of Pakistan also launched a bilingual (Urdu and English) chatbot to address common concerns related to Covid-19. This chatbot can be accessed through Facebook messenger and serves as a platform to provide information to the users about the common symptoms, risks and also information on how to get tested for Covid-19.[44] In Mumbai, Bharat Petroleum Corporation Limited (BPCL) launched a bilingual chatbot, Urja, in order help customers with queries and issues more efficiently. To ensure it is inclusive, they have trained Urja on 13 languages and one of them is Urdu. [13]

### 2.3.1 Urdu Natural Language Processing

One of the most important parts of building a chatbot is creating a suitable NLU pipeline which can help the chatbot classify intents quickly and accurately. Since, our project involves building a chatbot in the Urdu language, we need to build an Urdu based NLP model. Natural language processing in any language requires large amounts of data for it to provide effective results [52]. Urdu being a scarce resource language, it is difficult to create an effective Urdu based NLP model [40]. When compared to Western languages, core linguistic resources such as corpora, tagsets, WordNet, and etc. are not readily available, and if they are, they are subject to licensing restrictions [31]. Natural Language Processing is a multi-part process. Following are few of the steps we will be using in our Urdu based NLP pipeline:

- Tokenization: Tokenization (aka word segmentation) is the task of splitting a document into small segments of useful information, often called tokens. Despite being the simplest of all techniques, it is one of the most important and obligatory techniques in NLP tasks [19]. Due to space between each word in the English language the task of detecting word boundaries is easier and helps in tokenization, with exceptions in some cases which can dealt by making rules. However, Urdu is morphologically rich language with different nature of its characters. Due to inconsistent usage of white space between words, text tokenization and sentence boundary disambiguation in Urdu becomes difficult as compared to other languages like English. Information of morphology, syntax, statistical analysis and semantics of the language needs to be taken into account in case of Urdu text tokenization [18].

- Part of Speech (POS) Tagging: Parts of Speech tagging refers to tagging and categorizing words in correspondence with a particular part of speech. The categorization of the words depends on both the definition of the word and the context in which the word is being used in. POS tagging helps NLP systems to extract more linguistic information, making them robust and more accurate [18]. There are three main type of approaches used for POS tagging: rule-based approaches, statistical or machine learning approaches and hybrid approaches. Urdu POS taggers also use these approaches [31]. The first Urdu POS tagger used a rule-based approach and was developed by Andrew Hardie [25]. The resultant tagset uses Urdu grammatical rules as defined by Schmidt with the EAGLE guideline morpho-syntactic annotation of the dataset. Hardie's tagger used 270 linguistic rules and was able to achieve an accuracy of 90% [25]. One of the main downsides of rule-based approaches is that the construction of rules for a large dataset is an ardous task [31]. Therefore, machine learning or hybrid approaches are considered over rule-bassed approaches. [7] implements one such machine learning approach using n-gram Markov models. They used unigram, bigram and Backoff models and were able to achieve an accuracy of 95% on the Backoff model.

- Parsing: Parsing in NLP refers to extracting syntactic structure of a text by analyzing the words of the text based on grammar of the language. Parsing helps in the natural understanding of the language and is considered one of the core components in NLP systems [28]. Similar to POS tagging, there are rule-based and machine learning approaches for parsing. [29] presents a rule-based approach for parsing Urdu text, which uses two pass parsing technique in order to assess the grammar of Urdu text. The model uses some base Phrase Structure Grammar (PSG) Rules to parse the sentence and in case of a failure, Movement Rules are applied and sentence is reparsed.

- Named Entity Recognition (NER): Named Entity Recognition is the task of classifying named entities in a text. Named entities refers to proper nouns in a text and the purpose of NER is to extract them and classify them according to their defined types [6]. There have been various NER models proposed for European languages such as English, and also for non-European languages such as Arabic, Persian, and South Asian languages. However, NER in Urdu is still in its initial phases [40]. NER models highly depends on extrinsic linguistic resources such as annotated corpora, human-made dictionaries, and gazetteers to improve its accuracy, however, the Urdu language, for now lack these resources [31]. One more factor that contributes to the complexity of building NER models in the Urdu language, is that it doesn't support capitalization. An NER system can be rule-based, statistical, or a combination of the two.

A rule-based NER system tags a corpus with named entities using pre-written rules. A statistical NER system uses a training dataset to learn the probabilities of named entities, whereas hybrid systems employ both [21]. However, rule-based approach results are more accurate than statistical techniques in the context of Urdu NER tasks [52].

## 2.4 Natural Language Generation Through Deep Learning Models

Before transformers and attention mechanisms were common for natural language generation, state-of-the-art architectures such as Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) were used for this task [22]. However, one of the main problems with these models was the vanishing gradient problem due to their deep architecture. Transformer-based language models marked a new era of advancements in natural language generation tasks. The GPT-2 [47] model, a 1.5 billion parameter transformer, was introduced in an effort to perform unsupervised learning for natural language processing tasks such as question answering, machine translation, reading comprehension, and summarization. It was trained on a dataset of 8 million web pages. Following this, many models based on GPT-2 were introduced. DLGNet [43], a transformer-based model is an extension of GPT-2 for multi-turn dialogue modelling. It proposes a hybrid of long-range transformer design and random informative padding injection.

Currently, there exist several pre-trained models for open-ended conversational agents. Meena [3], uses social media comments to perform pre-training for dialogue generation. Its neural network with 2.6 billion parameters has been trained to reduce the perplexity of every next token. Blender [54] suggests fine-tuning the pre-trained dialogue model with human-annotated datasets in order to emphasize the conversational traits of engagingness, knowledge, empathy, and personality.

While much work on natural language generation is being done in prominent languages such as English, many resource-poor languages including Urdu have been left behind due to the lack of relevant resources. There have been instances of work being done on dataset compilation and natural language generation in languages other than English [39, 23, 2, 17, 5]. However, to our knowledge, no such effort has been made in the Urdu language.

# 3. Software Requirement Specification (SRS)

## 3.1 Functional Requirements

- Home Page
  - The home page will contain a navigation sidebar with buttons to navigate to other options in the app.
    * Navigate to a visual assistant which provides buttons for basic daily actions like accessing user's playlist, calling favourite contacts, checking appointments etc.
    * The user is able to update profile and add any changes to their current information.
    * The user is able to access their daily schedule.
    * It will allow the user to contact support if facing any issues with the app.
    * Allow the user to logout from the app.
    * A panic button to alert the caretaker in case of emergencies.
  - Chat-bot
    * The home page will contain a voice button and message editor so user can input their queries for the chat-bot. The chat-bot will be able to:
    * Converse in spoken Urdu with the user
    * Answer user queries and perform actions like send a message, make a call etc.
  - The vertical ellipsis button will allow user to access the settings of the app.

- Schedule Page
  - The page will display all the created reminders (day-wise) including medicine/appointment reminders.
  - The page will allow user add new reminders and edit/delete the existing reminders.
- Login Page
  - The login page will ask for user name and password to login the app.
  - The login page will allow user to go to sign up page if they have not logged in before.
- Sign Up Page
  - The page will ask for the users details and profile info.
  - The page asks for the users preferences. They have the options to:
    * Opt for meal reminders
    * Nominate a caretaker and enter their info.
    * Set medicine reminders and timings.
    * Set YouTube playlists that the assistant will play for the user.
    * Allow or disallow the application from performing sentiment analysis.
- Visual Assistant Page:
  - The page will give the users a visual interface to perform basic functions such as calling a person, sending a text message, getting news updates or opening a mobile application.
  - The page will contain buttons with visual icons and Urdu text for all the above mentioned features.
- Administrative Functions (Test phase only):
  - The admin will be able to browse and view all the conversation history. (the functionality is provided through Rasa-X)
  - The admin will be able to add, edit and delete intents to optimize the chatbot. (the functionality is provided through Rasa-X)
  - When our application is live we will be storing chat history via encryption.

## 3.2 Non-functional Requirements

- Usability
  - The interface of the system should be user-friendly from the perspective of the elderly. Minimal effort should be required to learn and operate the application.
  - It should be easy to navigate, different functions should be easily accessible and used with a minimal number of tabs and sections.
- Performance
  - The front-end should display conversations, pages and should carry out tasks with minimum possible response time.
  - The speech-to-text and text-to-speech libraries should perform quick conversions with good accuracy.
- Portability
  - The system should work on Android Operating System and should cater to different mobile screen sizes and aspect ratios.
- Efficiency and Maintainability
  - Page loads and custom actions should be returned and formatted in a timely fashion.
  - The system should be modular. There should be as many reusable components and code replication should be avoided.
- Safety and Security
  - The system should be protected from unauthorized access to the stored data.
  - User information and conversation should be secured by using appropriate encryption.
  - User information such as audio and chat will not be used for analysis without their consent.
  - There should be two clearly defined roles of the users. These are 'users' and 'administrator'. Every user will be required to register if they want to utilize app features such as chatbot, medical reminders, etc.

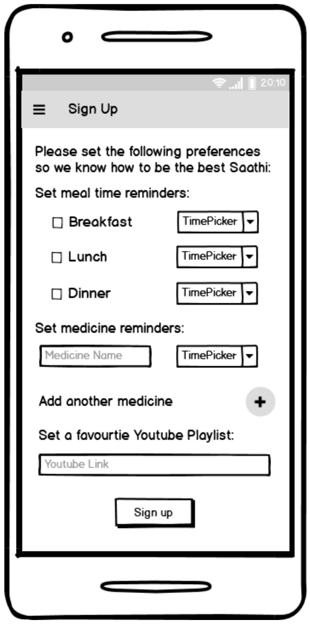
- The administrator will be allowed to improve chatbot performance using RASA X (during the testing phase) as well as the integration of new modules (if any) to the front-end.
- Appropriate authentication procedure should be followed for user login/ signup. The system should protect the user's account and data integrity should be maintained at all times.
- Administrators can only perform administrative tasks on pages they are privileged to access. Users will not be allowed to access the administrator pages.

## 3.3 External Interfaces

### 3.3.1 User Interfaces

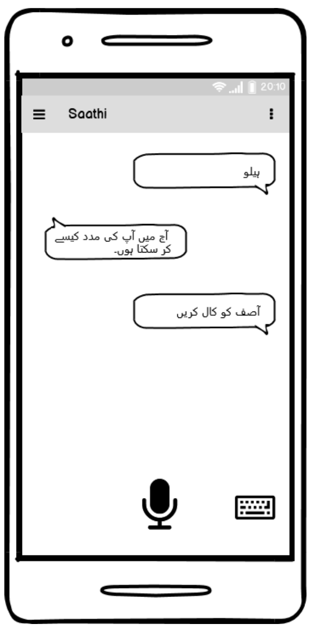

Login Page


Sign Up Page

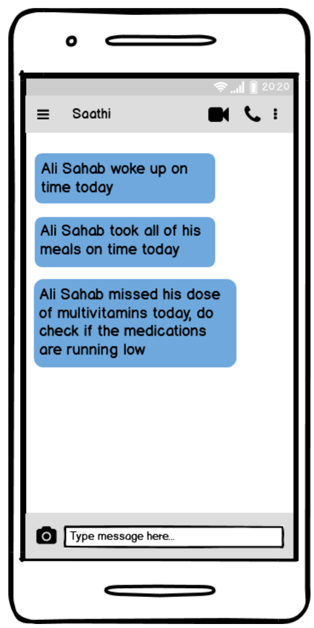



Preferences Page

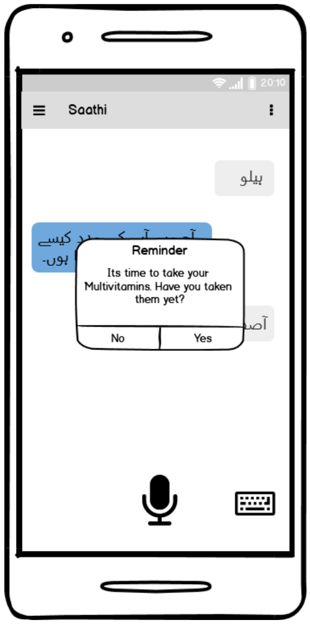

Caretaker Info Page



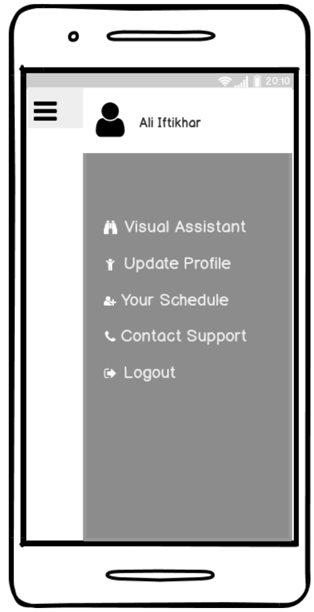

Homescreen with Chatbot




WhatsApp Notifications for Caretaker

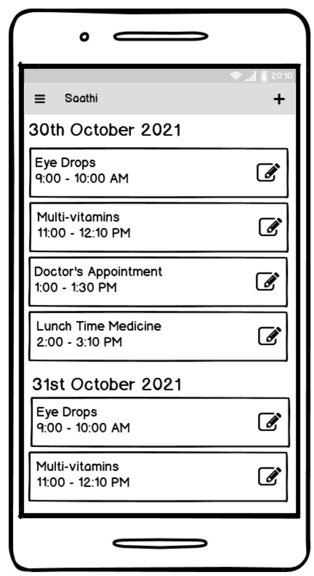


Medicine Reminder



Navigation Bar

Visual Assistant



Your Schedule Screen

### 3.3.2 Application Program Interface (API)

1. RASA: An open source machine learning framework for automated text and voice-based conversations (will be used for our chatbot).

2. Flutter: An open-source UI software development kit based on Dart language (Will be used to build the front-end of our application and for interaction with the back-end).

3. Twilio WhatsApp API: Will be used to send WhatsApp messages to the caretaker.

4. Restful APIs: We'll be using RESTful APIs to connect our chatbot to our application.

### 3.3.3 Hardware/Communication Interfaces

For the purpose of fine-tuning the Urdu natural language generation model, we used the **NVIDIA TITAN RTX TU102** GPU which was provided by the university.

## 3.4 Use Cases


Figure 3.1: Use Case Diagram

## 3.5 Datasets

### 3.5.1 Universal Dependencies Dataset

Universal Dependencies [59] is an open-source project with the aim to produce cross-linguistically consistent treebank annotations for the development Natural Language Processing applications. The dataset contains treebanks for more than 100 languages including Urdu. The Urdu dataset contains 138,077 tokens extracted from 5130 sentences. The dataset breaks the sentences into tokens. These tokens are further tagged with parts of speech and morphological features such as modal auxiliaries and phase verbs. Urdu dataset uses all the 17 universal POS tags defined in [1]. However, it does not contain named entity tags.

### 3.5.2 Urdu Named Entity Recognition Dataset

The dataset is an open-source dataset available on Github [8]. It contains 132530 Urdu sentences tagged with named entities in the BILUO format. In BILUO format each token is tagged according to the following format:

- B - Token is the beginning of a multi-token entity.
- I – Token is inside a multi-token entity.
- L – Token is the last token of a multi-token entity.
- U – Token is a single-token unit entity.
- O – Token is outside an entity.

```
[('ایل','B-Organization'),('ڈی','I-Organization'),('اے','L-Organization'),('پلازہ','O'),('آتشزدگی','O')]
[('لڑکا','O'),('نہاتے','O'),('ہوئے','O'),('راوی','O'),('میں','O'),('ڈوب','O'),('گیا','O')]
[('75','O'),('سالہ','O'),('نامعلوم','O'),('شخص','O'),('کی','O'),('پراسرار','O'),('ہلاکت','O')]
[('جہانزیب','B-Person'),('اعوان','L-Person'),('لاہور','U-Location'),('پ','O')]
```

Figure 3.2: Screenshot from the Urdu Named Entity Recognition Dataset

### 3.5.3 Urdu Dataset for RASA Chatbot

We built our own dataset for RASA chatbot which contains all the intents, entities and actions in order to respond to queries by the elderly while using our chatbot.

### 3.5.4 Dataset Compilation for Conversation Generation

The task of conversation generation requires a dataset of conversations in Urdu for training purposes. An example of input would be a statement in Urdu made by a speaker that describes any personal experience. The corresponding output would then be a reply to this statement that is grammatically and contextually sound and empathetic to the initial speaker's feelings. The models are trained on these conversation pairs to generate appropriate responses in Urdu. However, no such datasets are currently available for the Urdu language. Hence, two open-source English language datasets were used and processed for this task. The preparation of the UrduSaathi Dataset consisted of two main parts: firstly, both open-source datasets were preprocessed and machine translated into Urdu. Then in the second step, a subset of the dataset was manually corrected, and together, both formed the final dataset. A detailed description of the preparation of the dataset is as follows.

The UrduSaathi Dataset consists of two open-source datasets, the EmpatheticDialogues Dataset [50] and the ChitChat Dataset [41]. Both of these datasets contain multi-turn short conversations on various topics that mimic real human-like conversations in the English language.

The EmpatheticDialogues Dataset is a large-scale multi-turn empathetic dialogue dataset containing 24,850 one-to-one open-domain conversations. Each conversation was obtained by pairing two people together: a speaker and a listener. The speaker was asked to talk about some personal experience while the listener responded empathetically after inferring the underlying emotion from the speaker's words [50]. The context (underlying emotions) and self-evaluation fields, from the original dataset, were removed from the UrduSaathi dataset as they were unnecessary for our approach.

The ChitChat Dataset [41] is an open-domain conversational dataset from the BYU Perception, Control & Cognition lab's Chit-Chat Challenge. It contained multi-turn dialogues between two individuals that discussed a wide range of topics and usual small talks. These dialogue pairs helped the models generalize over different ways one can respond to common chitchat questions and dialogues.

After initial preprocessing and data-cleaning, Google Translate was used to machine translate these existing datasets into the Urdu language. Due to machine translation limitations in resource-poor languages like Urdu, the machine-translated dataset was found to have abundant syntactic and grammatical mistakes. For this reason, manual correction of the translations of a subset of the machine-translated dataset was performed for improvements in the grammar and sentence structure of the generated responses. The advantage of manual translation is near perfect precision; however, the task is highly labor-intensive. Therefore, the whole ChitChat Dataset (1492 utterances) and around 1120 utterances of the EmpatheticDialogues Dataset were successfully manually translated. After concatenating both machine-translated and manually corrected parts of the two datasets, a consolidated dataset of 77548 utterances was obtained. Figure 3.3 shows a complete conversation example from the consolidated dataset, where every conversation is stored in the form of dialogues.

The XGLM model required a list of entire conversations to be sent to the model with appropriate separators as input for training. The mT5 model required question-answer pairs for model fine-tuning. Hence, the dialogues were stored in a question and response pair at each line for this purpose. At the same time, the DialoGPT model required the context of the past utterances and the current response. For this paper, the number of past utterances was set to three.

| | |
|---|---|
| User 1: I broke my phone. | میں نے اپنا فون توڑ دیا۔ |
| User 2: Oh I am sorry to hear that, how? | اوہ مجھے یہ سن کر افسوس ہوا، کیسے؟ |
| User 1: I dropped it lol. | میں نے اسے گرا دیا۔ |
| User 2: Oh no! I hope you can fix it! | ارے نہیں! مجھے امید ہے<br>کہ آپ اسے ٹھیک کر سکتے ہیں! |

Figure 3.3: A conversation example from the UrduSaathi Dataset

## 3.6 System Diagram

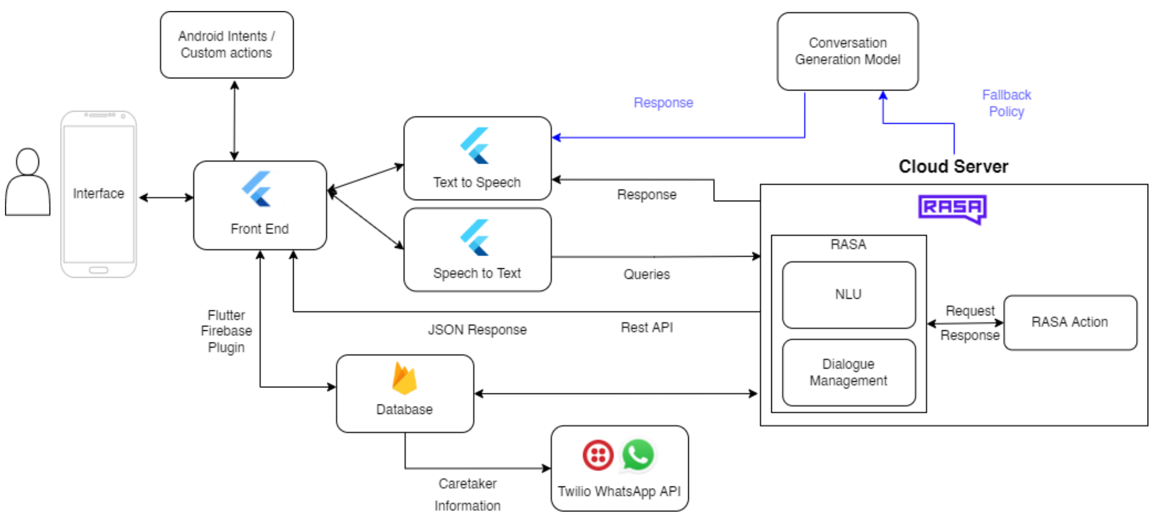



Figure 3.4: System Diagram

# 4. Software Design Specification (SDS)

## 4.1 Software Design

The UML class diagram shows the different screens that will be a part of our application. Since the app is user-specific, the user will first have to register using the registration screen and every time a new user object is created, their medical information and preferences will be saved in order to make sure they are given relevant reminders throughout their day. Once they have registered and logged in, they will be redirected to the chat screen where the user will be able to interact with the chatbot. Additionally, there is a visual virtual assistant screen that can be used to perform basic everyday actions like calling, taking pictures, playing their YouTube playlist, setting a reminder etc. via one click buttons. The caretaker's phone number will be used to send them message on WhatsApp about the elderly user's daily updates. This will be done using the message scheduler class and Twilio API. The user is connected to various classes like user log class which is used to store the information that needs to be sent to the caretaker at the end of the day. Figure 4.1 represents the UML structure of our application. A high quality image of the class diagram can be found here.

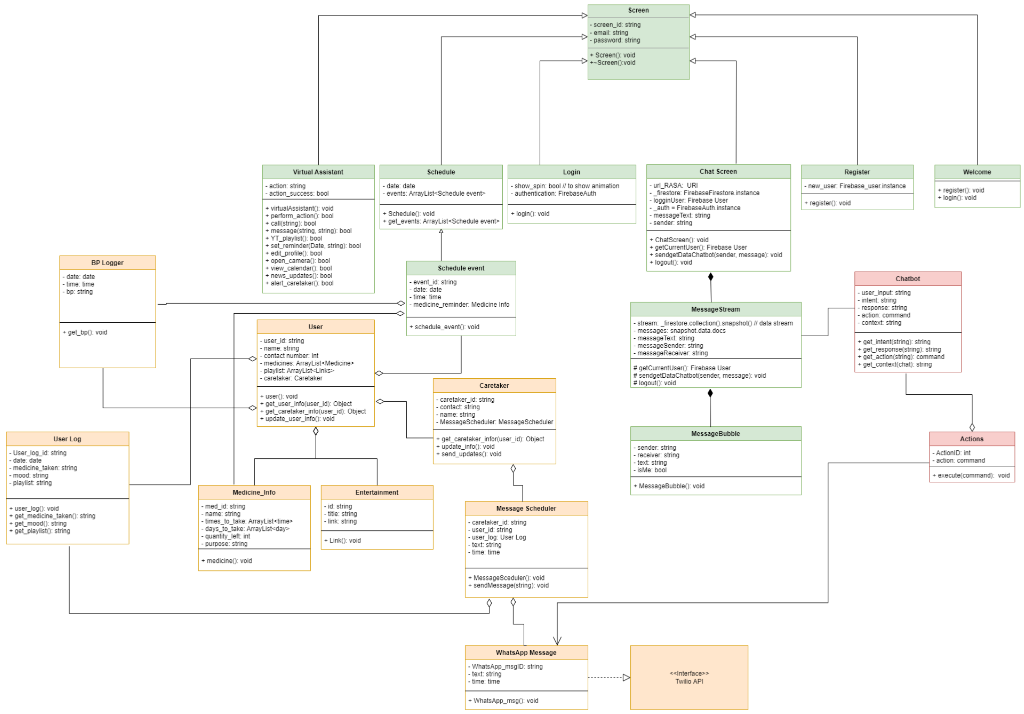



Figure 4.1: UML class diagram of the system.

## 4.2 Data Design

This section presents the structure of our database that caters to persistent data storage in our project. Figure 4.2 shows our ERD and a high quality image of the class diagram can be also found at here.

### 4.2.1 Data Pipeline

#### Data Pipeline for Chatbot Framework (RASA)

The input to the data pipeline is the Urdu text converted from the user input audio. RASA framework breaks down the processing of user query and returning of response into two components. The text input is first passed through the NLU component of the framework which involves natural language processing over the raw text and outputs intents and entities. There are two parts of the NLU component, 1) Urdu NLP Pipeline and 2) Entities Extractor and Intent Classifier. The raw Urdu text is first passed into the NLP pipeline which outputs processed query. This processed query is then passed into the intent classifier and entity extractor, which maps it to a pre-defined intent and extracts important entities from it. These extracted intents and entities are then passed on to the second component of the framework called RASA core. RASA core uses a classifier which finds out the best response to the query and outputs the response, which is then sent to the user. Table 4.3 shows the complete data-pipeline for our chatbot module.

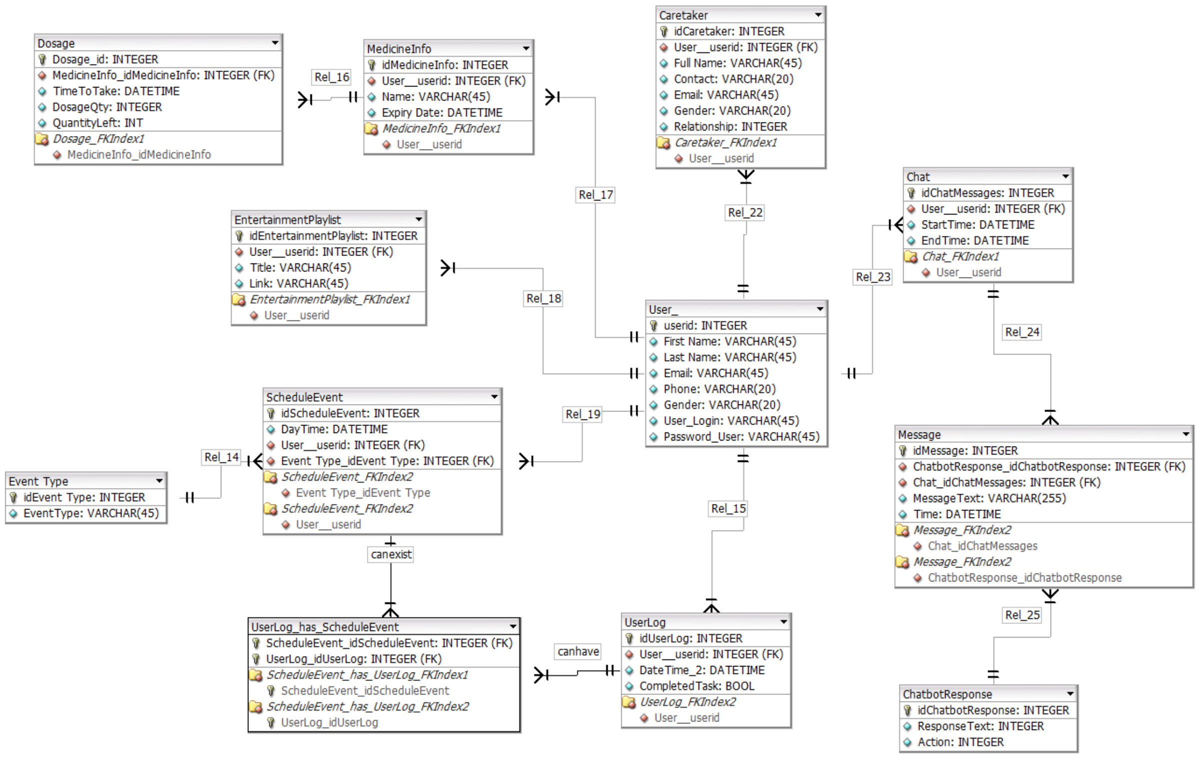



Figure 4.2: ERD Diagram for the database

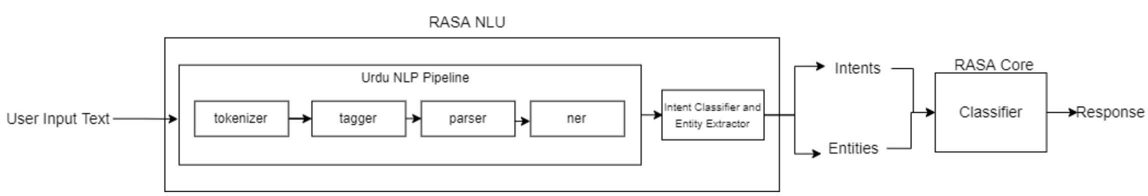



Figure 4.3: Data Pipeline for Chatbot Framework

## 4.3 Deep Learning NLG Models Used for Transfer Learning

For the task of conversation generation in Urdu, our work investigates the performance of three different pre-trained transformer-based models on the UrduSaathi Dataset. The three models are: DialoGPT [67] which is an English conversation generation model, and two multi-lingual models, mT5 [65], and XGLM [36].

Table 4.1 shows the values of the hyper-parameters used during the fine-tuning of these models.

### DialoGPT

DialoGPT (dialogue generative pre-trained transformer) is an open-source tunable gigaword scale neural network model for the generation of conversational responses for the English language [67]. DialoGPT extends GPT-2, which is a transformer-

Table 4.1: Hyper-parameter Values

| | DialoGPT | mT5 | XGLM |
|---|---|---|---|
| Optimizer | AdamW | Adafactor | AdamW |
| Learning rate | $5 \times 10^{-5}$ | $3 \times 10^{-4}$ | $2 \times 10^{-5}$ |
| Batch-size | 4 | 8 | 4 |
| Epochs | 15 | 7 | 4 |

based language model that was trained on 40GB of internet text [47]. It was then fine-tuned on Reddit conversations of 147M exchanges [67]. The DialoGPT-Medium model with 345M parameters and 24 layers was used in this work. This model was chosen as it has adequate performance in related tasks and is less computationally expensive than its larger counterpart [67].

For the fine-tuning process, the compiled dataset was first uploaded with the maximum input sequence length for the model set to the default value (128). Choosing the appropriate input sequence and batch size was necessary as long input sequence results in a slow training process, and a large batch size would cause memory overflow due to memory constraints of the GPU. Hence, there is a trade-off between training speed and memory constraints.

#### mT5

Google's T5 is a Text-To-Text Transfer Transformer model, which is a shared NLP framework where all NLP tasks are re-framed into a unified text-to-text format where the inputs and outputs are always text strings [48]. The T5 allows the use of the same model along with the loss function and hyper-parameters on any NLP task in English. For the purpose of this paper, the mT5 model, a multilingual variant of T5, was used. This model was pre-trained on mC4, a multilingual version of the C4 dataset comprising of 101 languages, including Urdu. For the fine-tuning process, the AdamW and Adafactor optimizers were tested, and the Adafactor optimizer performed better in the case of loss optimization.

#### XGLM

XGLM is a pre-trained multilingual large-scale auto-regressive model by Meta AI [36]. One of the main advantages of using this model is that it is trained on a large-scale balanced corpus - extracted from CommonCrawl (CC100-XL) - covering a diverse set of 30 languages, including Urdu. This allows it to outperform models like GPT-3 in multilingual reasoning and natural language inference, as GPT-3 has been trained on predominantly English training data [36]. XGLM consists of 4 different pre-trained models with varying parameters of up to 7.5 billion. For this paper, the model variant with 564M parameters with 24 layers was used, which was pre-trained on 30 languages.

# 5. Experiments and Results

## 5.1 Results for RASA Conversational Agent

Since the conversational agent will be directly interacting with elderly people it must be as accurate and well-performing as possible. To do this, the model was tested through the RASA test module. The cross-validation method was used to test the chatbot, where the dataset is split into k number of groups, each of equal size, in this case, $k = 5$. One of these $k$ splits is then chosen for testing and the rest of the splits are used for training the chatbot. This process is repeated $k$ number of times until each split is used for testing. Finally, a weighted average of all k iterations is taken to validate the model's performance. Figure 5.1 shows the confusion matrix of intents with the cross-validation approach. It shows that even though the diagonal values are high, there are several misclassifications of the intents. The ratio of misclassifications to correct classifications is 0.247. While the model performs well on a majority of the intents even with cross-validation, there is still room for improvement. Due to these misclassifications, a threshold of 0.7 is set as the confidence level of an intent classification. This implies that if the confidence score of any intent goes below the defined threshold value, a fallback policy is triggered and the user is asked to rephrase the sentence. The overall goal is to ensure that no inconvenience or discomfort is caused to the user by any incorrect or hurtful responses.

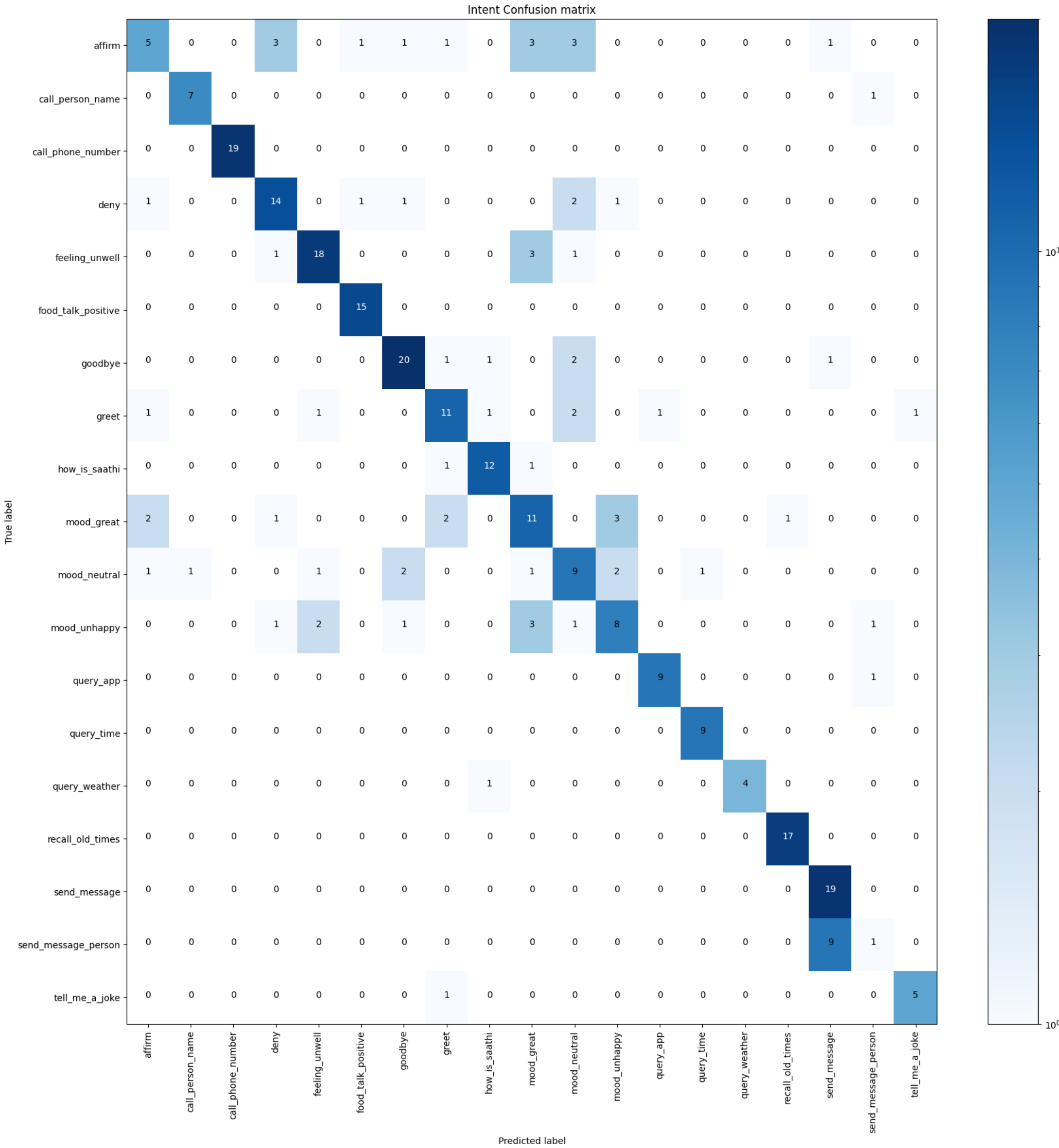



Figure 5.1: Confusion matrix of intents with cross-validation

## 5.2 Evaluation and Results for Conversation Generation

This section discusses the metrics used for evaluating the models and the results obtained.

### 5.2.1 Metrics & Experiments for Evaluation

In natural language generation, machine-generated text (output of the model) is usually evaluated against a target text (what is expected of the model to generate ideally). Therefore in order to measure the performances of our trained models, the right metrics identified for this task were BLEU [45], ROUGE [35], perplexity, and human evaluation. The BLEU and ROUGE metrics were used as they are widely reported metrics for conversation generation. They are precision and recall focused, which calculates the n-gram overlap of the reference and the generated texts. A perfect match results in a score of 1.0, whereas a perfect mismatch results in a score of 0.0. Along with BLEU and ROUGE, perplexity is used as a metric to evaluate the efficacy of generative models. By comparing the token distribution generated by the models to the one found in the test data, it measures how well the models have predicted the test dataset.

While these evaluation metrics are excellent at accounting for the evaluation of whole systems; they have been criticized for failing to gauge the quality of individual sentences [56] due to being uninformative and not suitable to assess the linguistic properties [33]. Therefore, there is a need to have a more fine-grained analysis of the results produced. This can be achieved through human evaluation of the test results on the basis of relevance, grammatical correctness, and human likeness. For this purpose, 30 native Urdu speakers were asked to assess the results of the models. The evaluators were shown various utterances and their corresponding responses generated by the DialoGPT, mT5, and XGLM models. They were then asked to evaluate each of the models' ability to reply in a human-like manner, its contextual relevance, and the grammatical correctness of the generated responses. The evaluators were asked to rate the responses as follows:

1. Human-likeness - rating the responses based on how natural they found the replies on a scale from 1 to 3 where 1 represented *not human-like*, 2 represented *somewhat human-like* and 3 represented *close to human-like* responses.

2. Relevance - rating how relevant the generated responses were in context to the input utterance on a scale from 1 to 3 where 1 represented *completely irrelevant*, 2 represented *somewhat relevant* and 3 represented *relevant* responses.

3. Grammatical correctness - rating how understandable and linguistically correct the response was on a scale from 1 to 3 where 1 represented *grammatically incorrect*, 2 represented *somewhat grammatically sound* and 3 represented *grammatically correct* responses.

### 5.2.2 Results

#### Automated Metrics

Table 5.1 shows the experimental results using automated metrics for the three models. The best perplexity score on the test dataset was achieved by the DialoGPT model, followed by the mT5 model, while the XGLM model had the worst score in terms of perplexity. If we consider BLEU Scores, the mT5 model outperformed the other two models for both BLUE-2 and BLEU-4 scores. DialoGPT achieved the worst score of 0.86% and 3.81% in BLUE-2 and BLEU-4 respectively. Finally, mT5 has the highest ROUGE score values among all the three models, followed by XGLM and DialoGPT.

The mT5 model achieved an overall good score in terms of ROUGE and BLEU. However, the DialoGPT model outperformed both mT5 and XGLM in terms of the perplexity score.

Table 5.1: Results of Automated Metrics of the fine-tuned models

| Method | ROUGE | | | BLEU | | Perplexity |
|---|---|---|---|---|---|---|
| | ROUGE-1 | ROUGE-2 | ROUGE-L | BLEU-2 | BLEU-4 | |
| DialoGPT (345M) | 11.63% | 1.43% | 10.47% | 3.81% | 0.86% | **1.37** |
| mT5 (580M) | **16.29%** | **2.80%** | **14.53%** | **6.12%** | **1.57%** | 7.49 |
| XGLM (564M) | 13.24% | 2.01% | 11.87% | 5.03% | 1.38% | 9.87 |

### Human Evaluation

The results of the survey have been presented in Table 5.2. The averaged scores for each metric are out of 3 as per the scale mentioned above. The results indicate that the mT5 model generates the most human-like natural responses with a high average score of 2.09 out of 3 in human-likeness. mT5 also outperforms DialoGPT and XGLM by achieving a substantially higher score of 2.04 for contextually relevant responses. For grammatical correctness, on the other hand, the XGLM model surpasses mT5 and DialoGPT by generating the most grammatically sound responses with an average score of 2.27.

Table 5.2: Results of Human Evaluation Survey

| Model | Human-likeness | Relevance | Grammatical Correctness |
|---|---|---|---|
| DialoGPT | 1.78 | 1.59 | 2.08 |
| mT5 | 2.09 | 2.04 | 2.17 |
| XGLM | 2.04 | 1.87 | 2.27 |

### Generated Responses

Figure 5.2 shows the generated responses from all three trained models on the same source input taken from the test dataset. The results obtained for this particular example were contextually, syntactically, and semantically sound. However, this is not the case for all the responses obtained.

For a dialogue generative model, it is imperative to produce grammatically and contextually sound responses. These responses should also have an overall human-likeness to them so that they can be utilized for the purpose of conversation generation over multiple applications. As is apparent from the results, the multilingual models: mT5 and XGLM have outperformed the DialoGPT model (pre-trained only on English data) on the human evaluation, BLEU, and ROUGE metrics. These

| | |
|---|---|
| Soure Input | میں آج بہت خوش ہوں کیوں کہ<br>میں نے گریجویشن کرلی ہے! |
| Translated Source Input | I am very happy today<br>because I have graduated! |
| DialoGPT Response | یہ بہت اچھا ہے! کیا آپ کو لگتا<br>ہے کہ آپ بہت خوش ہیں؟ |
| Translated DialoGPT Response | That is great! Do you think<br>that you are happy? |
| mT5 Response | مبارک ہو! آپ کو کیا ملا؟ |
| Translated mT5 Response | Congratulations! What did you get? |
| XGLM Response | مبارک ہو آپ نے بہت اچھا کیا<br>آپ کو اس پر فخر ہونا چاہیے۔ |
| Translated XGLM Response | Congratulations, you did so well<br>you should be proud of it. |

Figure 5.2: Responses generated from all three trained models

results are not surprising owing to the fact that both XGLM and mT5 have been pre-trained on large multilingual corpora that contained Urdu data.

For the multilingual models mT5 and XGLM, the obtained results show that the models are generating impressive results both in terms of grammatical correctness and human-likeness. From this, it can be inferred that these models successfully learn the underlying human-like conversation patterns in Urdu both semantically and syntactically. Figure 5.3 shows a response to an input using the mT5 model. Even though the generated response is syntactically and semantically correct, it can be observed that the response is irrelevant to the source input. This shows that the model has been unable to capture the true meaning of the input and has struggled to generate a relevant response. This may be due to the limitations of Urdu vocabulary of the model and the UrduSaathi dataset.

Similarly, Figure 5.4 shows a response generated by the DialoGPT model. The response is irrelevant as well as it contains grammatical and structural issues.

Therefore, it can be seen that although mT5 and XGLM have considerably good results as compared to DialoGPT, there are still limitations to how well the models perform. There is much room for improvement in generating fluent and contextual conversations. This can be further improved by enhancing the training dataset with more open-ended and grammatically correct Urdu conversations.

The differences in model performance between mT5 and XGLM can be attributed to a few factors. Firstly, the mT5-Base model has 580M parameters as compared to XGLM's 564M. This allows mT5 to learn more weights, enabling the model to generalize better on unseen data. Secondly, mT5 was pre-trained on 2.4B Urdu

| | |
|---|---|
| Source Input | میرا دانت ہل رہا ہے اور مجھے<br>لگتا ہے کہ یہ گر جائے گا۔ |
| Translated Source Input | My tooth is loose and<br>I think it will fall out. |
| Generated Response | یہ خوفناک ہے! کیا آپ کے<br>پاس کوئی پالتو جانور تھا؟ |
| Translated Generated Response | That is terrifying!<br>Did you have any pets? |

Figure 5.3: Generated sample from the mT5 model

| | |
|---|---|
| Source Input | میری آج ایک بزنس میٹنگ ہے۔ |
| Translated Source Input | I have a business meeting today. |
| Generated Response | یہ بہت اچھا ہے! کیا آپ<br>کے پاس کوئی بہن تھا؟ |
| Translated Generated Response | This is great!<br>Did you have a sister? |

Figure 5.4: Generated sample from the DialoGPT model

tokens giving it a significant advantage over XGLM, which was pre-trained on 1.35B Urdu Tokens. Thus it can be inferred that the size of the dataset used for pre-training of the mT5 model gives it leverage over XGLM in terms of performance and generating good results.

One of the main reasons for grammatical incorrectness in the case of DialoGPT is the lack of pre-training on the Urdu dataset and its sole reliance on the usage of machine-translated sentences in the compiled dataset used for training. The machine-translated sentences help the model understand the underlying context, however, due to improper structure and grammar of the input sentences themselves, the model struggles to generate more human-like responses that are also grammatically correct.

# 6. Conclusion and Future Work

We presented a one of its kind Urdu language based virtual assistant named "Saathi", in the form of a mobile application. It can help the Urdu-speaking elderly individuals navigate through their smartphones with ease and provide them with a supportive environment to stay entertained and engaged in daily activities. Saathi serves as a companion to the elderly and helps them stay connected with their loved ones.

Our work further discussed the limitations of resources for Urdu in terms of the conversational dataset as well as available open-domain conversation systems. In contrast to languages like English which have large corpora available for language generation along with large-scale pre-trained models with millions of parameters like GPT-2 and DialoGPT, Urdu is a resource-poor language with very limited datasets for conversation generation and pre-trained models for open-domain conversations.

As a result, we introduced the UrduSaathi Dataset, which contains both machine and manual translations of two open-source datasets, namely, the ChitChat Dataset [41] and the EmphatheticDialogue Dataset [50]. We investigated the fine-tuning of three large scale models, DialoGPT [67], mT5 [65] and XGLM [36] on the compiled dataset. We assessed the responses obtained from each of these models based on four metrics: BLEU score, ROUGE score, perplexity, and human evaluation. It was found that mT5 outperformed the rest in most of the automated metrics as well as human evaluation. Overall, the obtained results showed the advantage of using fine-tuning of these large-scale models on the compiled dataset. The future work aims to increase the size of the UrduSaathi Dataset with more open domain conversations in order to improve the performance of the pre-trained models.

# Appendix A. Data

1. The forms used for Human Evaluation for conversation generation models:
   (a) XGLM - A
   (b) XGLM - B
   (c) DialoGPT - A
   (d) DialoGPT - B
   (e) mT5 - A
   (f) mT5 - B
2. Survey for information gathering regarding the application requirements: Saathi - Urdu Virtual Assistant for Elderly

# References


[1] URL: https://universaldependencies.org/u/pos/.

[2] Tosin P. Adewumi et al. "Småprat: DialoGPT for Natural Language Generation of Swedish Dialogue by Transfer Learning". In: *CoRR* abs/2110.06273 (2021). arXiv: 2110.06273. URL: https://arxiv.org/abs/2110.06273.

[3] Daniel Adiwardana et al. "Towards a human-like open-domain chatbot". In: *arXiv preprint arXiv:2001.09977* (2020).

[4] *Ageing population in Pakistan*. URL: https://ageingasia.org/ageing-population-pakistan/ (visited on 09/17/2021).

[5] Kaveri Anuranjana, Anvesh Rao Vijjini, and Radhika Mamidi. "HindiRC: A Dataset for Reading Comprehension in Hindi". In: Apr. 2019.

[6] Waqas Anwar, Xuan Wang, and Xiao-long Wang. "A Survey of Automatic Urdu Language Processing". In: *2006 International Conference on Machine Learning and Cybernetics*. 2006, pp. 4489–4494. DOI: 10.1109/ICMLC.2006.259164.

[7] Waqas Anwar et al. "A Statistical Based Part of Speech Tagger for Urdu Language". In: *2007 International Conference on Machine Learning and Cybernetics*. Vol. 6. 2007, pp. 3418–3424. DOI: 10.1109/ICMLC.2007.4370739.

[8] Arbasit. $Arbasit/Urdu_ner : UrduNerUsingLSTM$. URL: https://github.com/arbasit/Urdu_NER.

[9] *Aurat Raaj*. URL: https://auratraaj.co/ (visited on 09/17/2021).

[10] Jonas Beskow et al. "The MonAMI Reminder: a spoken dialogue system for face-to-face interaction". In: *Tenth Annual Conference of the International Speech Communication Association*. Citeseer. 2009.

[11] Timothy W. Bickmore et al. "A Randomized Controlled Trial of an Automated Exercise Coach for Older Adults". In: *Journal of the American Geriatrics Society* 61.10 (2013), pp. 1676–1683. DOI: https://doi.org/10.1111/jgs.12449. eprint: https://agsjournals.onlinelibrary.wiley.com/doi/pdf/10.1111/jgs.12449. URL: https://agsjournals.onlinelibrary.wiley.com/doi/abs/10.1111/jgs.12449.

[12] James A Blumenthal et al. "Effects of exercise training on older patients with major depression". In: *Archives of internal medicine* 159.19 (1999), pp. 2349–2356.

[13] *BPCL launches AI-enabled chatbot Urja*. URL: https://www.livemint.com/companies/news/bpcl-launches-ai-enabled-chatbot-urja-11629439490779.html (visited on 09/17/2021).

[14] J. Cassell et al. "Requirements for an Architecture for Embodied Conversational Characters". In: *Eurographics Computer Animation and Simulation '99* (1999), pp. 109–120. DOI: 10.1007/978-3-7091-6423-5_11.

[15] Iv Com. *Global strategy and action plan on ageing and health*. 2017. ISBN: 9789241513500. URL: http://apps.who.int/bookorders..

[16] Angelo Costa et al. "PHAROS—PHysical assistant RObot system". In: *Sensors (Switzerland)* 18.8 (2018), pp. 1–19. ISSN: 14248220. DOI: 10.3390/s18082633.

[17] Martin d'Hoffschmidt et al. "FQuAD: French Question Answering Dataset". In: *CoRR* abs/2002.06071 (2020). arXiv: 2002.06071. URL: https://arxiv.org/abs/2002.06071.

[18] Ali Daud, Wahab Khan, and Dunren Che. "Urdu language processing: a survey". In: *Artificial Intelligence Review* 47 (Mar. 2017). DOI: 10.1007/s10462-016-9482-x.

[19] Nadir Durrani and Sarmad Hussain. "Urdu Word Segmentation". In: *Human Language Technologies: The 2010 Annual Conference of the North American Chapter of the Association for Computational Linguistics*. HLT '10. Los Angeles, California: Association for Computational Linguistics, 2010, pp. 528–536. ISBN: 1932432655.

[20] Roos Eisma et al. "Mutual inspiration in the development of new technology for older people". In: (Jan. 2003).

[21] Karthik Gali et al. "Aggregating machine learning and rule based heuristics for named entity recognition". In: *Proceedings of the IJCNLP-08 Workshop on Named Entity Recognition for South and South East Asian Languages*. 2008.

[22] Albert Gatt and Emiel Krahmer. "Survey of the State of the Art in Natural Language Generation: Core tasks, applications and evaluation". In: *CoRR* abs/1703.09902 (2017). arXiv: 1703.09902. URL: http://arxiv.org/abs/1703.09902.

[23] Deepak Gupta, Asif Ekbal, and Pushpak Bhattacharyya. "A Deep Neural Network Framework for English Hindi Question Answering". In: *ACM Trans. Asian Low-Resour. Lang. Inf. Process.* 19.2 (Nov. 2019). ISSN: 2375-4699. DOI: 10.1145/3359988. URL: https://doi.org/10.1145/3359988.

[24] Francisco J. Gutierrez et al. "Assembling mass-market technology for the sake of wellbeing: a case study on the adoption of ambient intelligent systems by older adults living at home". In: *Journal of Ambient Intelligence and Humanized Computing* 10.6 (2017), pp. 2213–2233. DOI: 10.1007/s12652-017-0591-4.

[25] A Hardie. "Developing a tagset for automated part-of-speech tagging in Urdu." English. In: Corpus Linguistics 2003 ; Conference date: 01-03-2003. 2003.

[26] *HOBBIT Project FP7*. URL: http://hobbit.acin.tuwien.ac.at/ (visited on 09/17/2021).

[27] Minlie Huang, Xiaoyan Zhu, and Jianfeng Gao. "Challenges in building intelligent open-domain dialog systems". In: *ACM Transactions on Information Systems (TOIS)* 38.3 (2020), pp. 1–32.

[28] S.M. Jafar Rizvi, M. Husssain, and N. Qaiser. "Language Oriented Parsing Through Morphologically Closed Word Classes in Urdu". In: *Student Conference On Engineering, Sciences and Technology.* 2004, pp. 19–24. DOI: 10.1109/SCONES.2004.1564762.

[29] H. Kabir. "Two-pass parsing implementation for an Urdu Grammar Checker". In: Feb. 2002, pp. 51–51. ISBN: 0-7803-7715-X. DOI: 10.1109/INMIC.2002.1310158.

[30] Zerrin Kasap et al. "Making them remember—Emotional virtual characters with memory". In: *IEEE Computer Graphics and Applications* 29.2 (2009), pp. 20–29.

[31] Wahab Khan et al. "Named Entity Dataset for Urdu Named Entity Recognition Task". In: (Jan. 2016).

[32] J. Klein, Y. Moon, and R.W. Picard. "This computer responds to user frustration: Theory, design, and results". In: *Interacting with Computers* 14.2 (Feb. 2002), pp. 119–140. ISSN: 0953-5438. DOI: 10.1016/S0953-5438(01)00053-4. eprint: https://academic.oup.com/iwc/article-pdf/14/2/119/2284802/iwc14-0119.pdf. URL: https://doi.org/10.1016/S0953-5438(01)00053-4.

[33] Chris van der Lee et al. "Human evaluation of automatically generated text: Current trends and best practice guidelines". In: *Computer Speech & Language* 67 (2021), p. 101151.

[34] S Lesin. "Frederick'medical virtual assistant". In: *Final project in Software Engineering Department* (2016).

[35] Chin-Yew Lin. "Rouge: A package for automatic evaluation of summaries". In: *Text summarization branches out.* 2004, pp. 74–81.

[36] Xi Victoria Lin et al. "Few-shot Learning with Multilingual Language Models". In: *CoRR* abs/2112.10668 (2021). arXiv: 2112.10668. URL: https://arxiv.org/abs/2112.10668.

[37] K. Crockett M. Kaleem J. O'Shea. "Development of UMAIR the Urdu Conversational Agent for Customer Service". In: *Proceedings of the World Congress on Engineering* (2014).

[38] Molly McLaughlin. *What Is a Virtual Assistant and How Does It Work?* Aug. 2021. URL: https://www.lifewire.com/virtual-assistants-4138533.

[39] Hussein Mozannar et al. "Neural Arabic Question Answering". In: *CoRR* abs/1906.05394 (2019). arXiv: 1906.05394. URL: http://arxiv.org/abs/1906.05394.

[40] Smruthi Mukund and Rohini K. Srihari. "NE Tagging for Urdu Based on Bootstrap POS Learning". In: *Proceedings of the Third International Workshop on Cross Lingual Information Access: Addressing the Information Need of Multilingual Societies.* CLIAWS3 '09. Boulder, Colorado: Association for Computational Linguistics, 2009, pp. 61–69. ISBN: 9781932432336.

[41] Will Myers, Tyler Etchart, and Nancy Fulda. "Conversational Scaffolding: An Analogy-Based Approach to Response Prioritization in Open-Domain Dialogs". In: (2020).

[42] *Ok Ameer Urdu and Hindi assistant for Android - APK Download*. May 2019. URL: `https://m.apkpure.com/ok-ameer-urdu-and-hindi-assistant/com.ameerhamza6733.okAmeer`.

[43] Oluwatobi Olabiyi and Erik T. Mueller. "Multi-turn Dialogue Response Generation with Autoregressive Transformer Models". In: *CoRR* abs/1908.01841 (2019). arXiv: `1908.01841`. URL: `http://arxiv.org/abs/1908.01841`.

[44] *Pakistan launches bilingual chatbot to address coronavirus concerns*. URL: `https://tribune.com.pk/story/2178577/8-pakistan-launches-bilingual-chatbot-address-coronavirus-concerns` (visited on 12/03/2021).

[45] Kishore Papineni et al. "Bleu: a Method for Automatic Evaluation of Machine Translation". In: *Proceedings of the 40th Annual Meeting of the Association for Computational Linguistics*. Philadelphia, Pennsylvania, USA: Association for Computational Linguistics, July 2002, pp. 311–318. DOI: `10.3115/1073083.1073135`. URL: `https://aclanthology.org/P02-1040`.

[46] *Population by 5 year age groups - Pakistan*. URL: `https://www.pbs.gov.pk/sites/default/files//tables/POPULATION%5C%20BY%5C%205%5C%20YEAR%5C%20AGE%5C%20GROUPS%5C%20-%5C%20PAKISTAN.pdf` (visited on 09/17/2021).

[47] Alec Radford et al. "Language models are unsupervised multitask learners". In: *OpenAI blog* 1.8 (2019), p. 9.

[48] Colin Raffel et al. "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer". In: *Journal of Machine Learning Research* 21.140 (2020), pp. 1–67. URL: `http://jmlr.org/papers/v21/20-074.html`.

[49] *RAMCIP — Robotic Assistant for MCI Patients at home*. URL: `https://ramcip-project.eu/` (visited on 09/17/2021).

[50] Hannah Rashkin et al. "Towards Empathetic Open-domain Conversation Models: A New Benchmark and Dataset". In: *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*. Florence, Italy: Association for Computational Linguistics, 2019, pp. 5370–5381. DOI: `10.18653/v1/P19-1534`. URL: `https://aclanthology.org/P19-1534`.

[51] Zobia Rehman, Waqas Anwar, and Usama Ijaz Bajwa. *Challenges in Urdu Text Tokenization and Sentence Boundary Disambiguation.*

[52] Kashif Riaz. "Rule-Based Named Entity Recognition in Urdu". In: *Proceedings of the 2010 Named Entities Workshop*. NEWS '10. Uppsala, Sweden: Association for Computational Linguistics, 2010, pp. 126–135. ISBN: 9781932432787.

[53] Lazlo Ring et al. "Social support agents for older adults: longitudinal affective computing in the home". In: *Journal on Multimodal User Interfaces* 9.1 (2014), pp. 79–88. DOI: `10.1007/s12193-014-0157-0`.

[54] Stephen Roller et al. “Recipes for building an open-domain chatbot”. In: *CoRR* abs/2004.13637 (2020). arXiv: 2004.13637. URL: https://arxiv.org/abs/2004.13637.

[55] Natasha Dow Schüll. “Data for life: Wearable technology and the design of self-care”. In: *BioSocieties* 11.3 (2016), pp. 317–333. DOI: 10.1057/biosoc.2015.47.

[56] Anastasia Shimorina. *Human vs Automatic Metrics: on the Importance of Correlation Design*. 2018. DOI: 10.48550/ARXIV.1805.11474. URL: https://arxiv.org/abs/1805.11474.

[57] Nirmalya Thakur and Chia Y Han. “An approach to analyze the social acceptance of virtual assistants by elderly people”. In: *Proceedings of the 8th International Conference on the Internet of Things*. 2018, pp. 1–6.

[58] Soe Ye Yint Tun, Samaneh Madanian, and Farhaan Mirza. “Internet of things (IoT) applications for elderly care: a reflective review”. In: *Aging Clinical and Experimental Research* 33.4 (2020), pp. 855–867. DOI: 10.1007/s40520-020-01545-9.

[59] *UD for Urdu*. URL: https://universaldependencies.org/ur/index.html.

[60] Ashiq Uzma and Asad Amir Zada. “The rising old age problem in Pakistan”. In: *Journal of the Research Society of Pakistan*. Vol. 54. 2017.

[61] *WHATSAPP CHATBOT ON COVID-19 IN URDU*. URL: https://www.thehindu.com/news/cities/Hyderabad/whatsapp-chatbot-on-covid-19-in-urdu/article31360026.ece (visited on 09/17/2021).

[62] “Why Population Aging Matters: A Global Perspective”. In: *PsycEXTRA Dataset* (2007). DOI: 10.1037/e610292007-001.

[63] *world population ageing 2019: highlights - united nations*. URL: https://www.un.org/en/development/desa/population/publications/pdf/ageing/WorldPopulationAgeing2019-Highlights.pdf.

[64] *World’s first Urdu based ‘Siri-styled’ voice-recognition AI bot launched*. URL: https://profit.pakistantoday.com.pk/2019/05/02/smart-cx19-focuses-customer-satisfaction-through-technology/ (visited on 09/17/2021).

[65] Linting Xue et al. “mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer”. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Online: Association for Computational Linguistics, June 2021, pp. 483–498. DOI: 10.18653/v1/2021.naacl-main.41. URL: https://aclanthology.org/2021.naacl-main.41.

[66] Kiyoshi Yasuda, Jun-ichi Aoe, and Masao Fuketa. “Development of an agent system for conversing with individuals with dementia”. In: *27 (2013)*. 2013, 3C1IOS1b2–3C1IOS1b2.

[67] Yizhe Zhang et al. “DialoGPT: Large-Scale Generative Pre-training for Conversational Response Generation”. In: *CoRR* abs/1911.00536 (2019). arXiv: 1911.00536. URL: http://arxiv.org/abs/1911.00536.